UNIVERSITÀ
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TRIDANT

STEP-BY-STEP INSTRUCTIONS

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SYSTEM ARCHITECTURE OVERVIEW

This document provides the technical blueprints and operational workflows to construct and validate our dual-band cylindrical DRA system and the proposed testing platform. The complete framework is divided into three functional blocks: a Transmission (TX) setup and two independent, interchangeable Reception (RX) characterization configurations.

TRANSMISSION (TX) SETUP

The transmission block generates the localized RF environment to excite distinct resonant modes inside the CDRA. It consists of:

- **HT008 Signal Generator**: Generates continuous-wave carriers at 1.575 GHz (GNSS) and 5.9 GHz (V2X). [Figure 2]
- **Branch-Line Hybrid Coupler**: A microstrip quadrature network that splits the 1.575 GHz signal into two outputs with a 90° electrical phase shift to generate circular polarization, fundamental for satellite communication.
- **Cylindrical Dielectric Resonator Antenna (CDRA)**: Instead of using standard metal wires, this antenna relies on a solid brick of ceramic-loaded material ($\varepsilon_r \approx 7.5$) to bounce and shape the radio waves.

RECEPTION (RX) SETUP A: HANDHELD POWER METER

A portable, battery-powered scanning device used to physically map out how the antenna distributes its energy in space.

- **HTOOL HT8 Log-Periodic Antenna:** A highly directional pointer antenna. Like a directional microphone that only picks up sound from where it is pointed, this antenna captures radio waves from a specific direction. [Figure 4]
- **AD8317 Logarithmic Detector:** A translator chip that converts high-frequency radio power into a simple DC voltage. The stronger the radio signal it catches, the lower the voltage it outputs, providing a highly predictable conversion scale. [Figure 1]
- **CodeCell C6 Microcontroller:** Acts as the brain and compass of the handle. The microcontroller samples the detector's voltage via its Analog-to-Digital Converter (**GPIO1**), while the onboard Inertial Measurement Unit (IMU) tracks the exact angular direction (yaw) of your sweep. It syncs these values and transmits them wirelessly via Bluetooth Low Energy (BLE) to a laptop. [Figure 3]

RECEPTION (RX) SETUP B: SDR SPECTRAL PLATFORM

A stationary, high-resolution setup used to verify the purity of the radio frequencies and check the alignment of the waves.

- **OpenSourceSDRLab 7020-SDR:** A Software-Defined Radio platform pairing a Xilinx Zynq-7020 processing chip with an Analog Devices AD9361 radio transceiver. It acts as a high-speed digital scanner, translating raw radio waves into real-time streams of digital data. [Figure 5]

- **SDR++ Software:** A computer application that processes the SDR data to generate live, interactive graphs (FFT spectrums) and color-coded "waterfall" displays. This allows you to visually inspect exactly how much signal is lost or maintained when you rotate the antennas.

ASSEMBLY AND CONSTRUCTION

DIELECTRIC RESONATOR ANTENNA FABRICATION

The cylindrical DRA is fabricated using Nanoe Zetamix Epsilon $\varepsilon_r = 7.5$ filament. Due to the mechanical and thermal characteristics of this particular filament it is important to follow specific configurations parameters for printing:

- **Printing Temperature**: 270°C – 300°C
- **Plate Temperature**: 100°C
- **Ambient Temperature**: 120°C
- **Infill Density**: 95%
- **Layer Thickness**: 0.1 mm
- **First Layer Speed**: 2.5 mm/s
- **General Printing Speed**: Slowed down to 5-7.5 mm/s
- **Cooling Fan**: 0%

If the fabrication setup lacks a controlled environment capable of maintaining the manufacturer's recommended 120°C ambient chamber temperature, it is advised to execute the following adhesion workarounds to prevent base distortion and detachment:

1. **Bed Preparation**: Apply a uniform layer of high-temperature resistant permanent glue stick across a flexible build plate.
2. **Slicer Adhesion Modification**: Program a wide **Custom Brim** rather than a standard skirt to securely anchor the perimeter of the cylinder base to the plate.
3. **Post-Print Cooldown Routing**: Do not remove the part immediately upon print completion. Program a manual, stepped cooldown of the heated bed to prevent severe thermal shock and subsequent geometric warping.

HANDHELD PORTABLE RF POWER METER

Wire the module according to the following hardware routing scheme:

- **RF Signal Input**: Connect the SMA output of the HTOOL HT8 log-periodic directional antenna directly to the RF input port of the AD8317 logarithmic power detector via a coax cable.
- **DC Signal Path**: Route the analog output voltage (V_{OUT}) pin of the AD8317 detector to GPIO01 (configured as an Analog Input pin) on the CodeCell C6 microcontroller platform.
- **Power Distribution**: Connect the terminals of the Lithium-ion battery to the input of the 5V step-up DC-DC converter module. Route the regulated 5V output rail of the converted to the V_{CC} supply pin of the AD8317 detector and the V_{IN} power pin of the CodeCell C6.

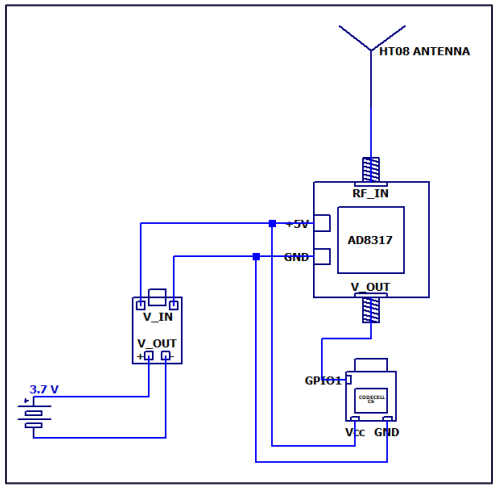


Figure 1 – Schematic of the handheld power meter.

OPERATION AND CALIBRATION WORKFLOW

CUSTOM HANDLE CALIBRATOIN AND FIELD SCANNING

The AD8317 outputs a negative-slope DC voltage linear-in-dB relative to the incident RF power level within its operating window of $-40\ dBm\ to\ -10dBm$. To convert the raw ADC voltage reading from GPIO01 into absolute power metrics (P_{IN}) the following transfer equation is used:

$$P_{IN} = \frac{V_{OUT} - V_{INTERCEPT}}{Slope}$$

To acquire the data using this system the following procedure needs to be followed:

1. Energize the HT008 signal generator and set the output to the target frequency (1.575 GHz for GNSS or 5.9GHz for V2X). Manual cable routing dictates whether the signal feeds the central probe directly or passes through the hybrid branch-line coupler network.
2. Power on the handle system by connecting the battery. The embedded firmware automatically initializes the onboard BNO086 IMU and activates the internal SAR ADC on GPIO01.
3. Execute the host Python script on the terminal to establish a Bluetooth Low Energy (BLE) connection with the CodeCell C6.
4. Stand at a fixed distance within the line-of-sight propagation path of the DRA. Hold the handle and, pointing the directional antenna at the DRA, rotate around it and complete a full 360° circle to capture data from all directions. The onboard firmware from the CodeCell C6 will sample the ADC voltage while the BLE radio is intermittently powered down, then briefly re-engage the transmitter to send synchronized pairs of orientation degrees (Yaw) and calibrated power values (P_{IN}) back to the laptop GUI.

SDR SPECTRUM AND WATERFALL POLARIZATION TESTING

This procedure visually validates the polarization characteristics of excited modes.

1. Launch SDR++ (or any preferred software) on the host PC and select the source backend driver corresponding to the SDR.

2. Set the VFO center frequency to match the active transmitter frequency (1.575 GHz or 5.9 GHz) and adjust the baseband gain stage to avoid digital clipping on the FFT display.
3. **Linear Polarization**
    I. Route the HT008 generator directly to one of the GNSS peripheral probes, omitting the hybrid coupler.
    II. Orient the receiving HT8 log-periodic antenna vertically to match the transmitted field polarization vector. Observe the signal peak level on the SDR++ FFT and the high-intensity trace on the waterfall.
    III. Rotate the receiving HT8 antenna physically by 90° into a horizontal plane. Observe the real-time attenuation drop on the waterfall display, validating linear cross-polarization isolation.
    IV. Repeat the process for the other peripheral probe, observing a similar but opposite behavior when rotating the antenna.
4. **Circular Polarization**
    I. Insert the hybrid branch-line coupler between the HT008 generator and the two orthogonal peripheral probes to enable quadrature feeding. It's important to connect a 50Ω load to port 4 (isolated).
    II. Monitor the SDR++ waterfall display while continuously rotating the receiving HT8 antenna through a slow 90° angular shift. Observe the baseline amplitude trace on the waterfall remain substantially constant throughout the full rotation sweep, confirming the generation of an symmetrical circularly polarized broadside field.

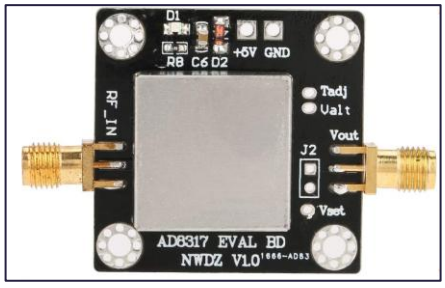
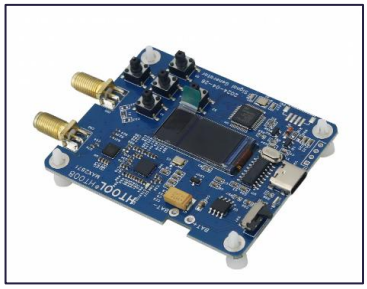


Figure 1: AD8317


Figure 2: HT008 Signal Generator

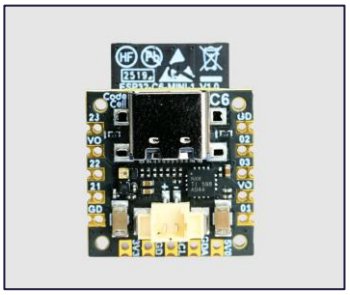


Figure 3: CodeCell C6



Figure 4: HT08 Log-Periodic Directional Antenna


Figure 5: Zynq-7020 SDR